

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 What is fading in a wireless communication channel? a) A permanent loss of signal power b) The variation in received signal strength due to environmental effects c) The process of signal amplification in antennas d) The interference caused by electromagnetic waves
- 2 What is the primary mode of ground wave propagation? a) Sky wave b) Space wave c) Surface wave d) Direct wave
- 3 What is the fundamental mechanism by which an antenna radiates electromagnetic waves? a) Conversion of mechanical energy into electromagnetic waves b) Oscillation of charges causing time-varying electric and magnetic fields c) Reflection of signals from the atmosphere d) Absorption of radio waves from space
- 4 What is the primary function of an antenna in a communication system? a) To generate power for signal transmission b) To convert electrical signals into electromagnetic waves for transmission and reception c) To store electromagnetic waves d) To block unwanted signals
- 5 What does the principle of pattern multiplication state in antenna arrays? a) The total radiation pattern is the direct sum of all individual elements b) The array radiation pattern is the product of the element pattern and array factor c) Each antenna element radiates independently without affecting the overall pattern d) The radiation pattern is unaffected by spacing between elements
- 6 What defines a broadside antenna array? a) The elements are arranged to radiate maximum energy along the axis of the array b) The elements are placed randomly without any directional preference c) The radiation is directed perpendicular to the array axis d) The array is designed to eliminate all side lobes
- 7 Which of the following best describes the analogy of a smart antenna system in wireless communication? a) A flashlight focusing light in all directions b) A loudspeaker broadcasting sound uniformly c) A person cupping their ears to hear a specific direction d) A mirror reflecting light randomly

- 8 . Which technique is commonly used in smart antennas to implement beamforming? a) Frequency hopping b) Time-division multiplexing c) Spatial filtering d) Amplitude modulation
- 9 What is the key characteristic of fixed weight beam forming in smart antennas? a) It dynamically adjusts weights based on signal environment b) It uses pre-determined weights that do not change c) It requires feedback from the receiver for operation d) It is only applicable to omnidirectional antennas
- 10 In a typical V-antenna configuration, what is the usual range of the apex angle between the two legs? a) 10° to 20° b) 36° to 72° c) 90° to 120° d) 150° to 180° .
- 11 What is the typical structural configuration of a microstrip (patch) antenna? a) A loop of wire suspended in air b) A flared waveguide with a conical shape c) A metallic patch on a dielectric substrate with a ground plane d) A vertical rod mounted above a ground plane.
- 12 What is the key characteristic of fixed weight beam forming in smart antennas? a) It dynamically adjusts weights based on signal environment b) It uses pre-determined weights that do not change c) It requires feedback from the receiver for operation d) It is only applicable to omnidirectional antennas
- Q2. Solve the following.
- A) Explain the term Virtual height, Maximum Usable Frequency (MUF) & Skip Distance 6
- B) Explain in brief: a) Ground wave propagation b) Space wave propagation 6
- Q3. Solve the following.
- A) Explore the various types of antenna 6
- B) Define the terms: 1) Bandwidth of Antenna 2) Input Impedance 3) Antenna Radiation Efficiency 6
- Q4. Solve Any Two of the following.
- A) Compare the Characteristics of a Binomial & Dolph Tschubyshev antenna array 6
- B) Explain the concept of Broad side array 6
- C) Write short notes on Log Periodic antenna 6
- Q5. Solve Any Two of the following.
- A) Explain the drawbacks of Smart antenna 6
- B) Explain different challenges associated with the implementation of Smart antenna in communication system 6
- C) Explain Antenna Beam forming 6
- Q6. Solve Any Two of the following.
- A) Explain the Microstrip antenna 6
- B) Write a short note on Helical antenna 6
- C) Explain the structural characteristics & dimensions of antennas with parabolic reflectors 6